Accept, Repeat or Confirm?

Confidence-Calibrated Clarification for Speech Slot Filling on a Service Robot

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Issue #1384 Data: TTS voices transcribed by the robot Whisper (proxy)

TTS proxy data. The audio is synthetic speech (edge-tts voices with 16 English accents, plus added noise and reverb), but every word probability comes from the robot STT (`distil-large-v3`, same decoding settings). The extraction LLM is emulated by rules. The numbers are a proxy for the week-2 human recordings, not a replacement: TTS voices are more regular than people. Re-run `run_all.py --data data` with the human sessions before shipping thresholds.

Abstract

FRIDA asks for a guest's name and favorite drink in the Human-Robot Interaction Challenge (HRIC). Before this work, a fixed heuristic decided whether to accept the transcribed value or ask "Did you say X?": both the Whisper word probability and the option similarity had to exceed 0.7. The thresholds had never been measured, and the 2027 rulebook rewards not asking such non-essential questions (+15) while penalizing every wrong memorized value (−40). We instrument the dialog, fix a matching bug that set the confidence of normalized values to zero, and replace the rule with a calibrated estimate of p(correct) that drives a three-way policy: *accept*, *ask to repeat* (free under the rules) or *confirm*. On held-out speakers, the selected model (Logistic all + isotonic) reaches AUROC 0.83 [0.74, 0.92] and ECE 0.090. The three-way policy changes confirmation questions by 78.5% relative to the current heuristic, changes the wrong-value rate by -6.8 pp, and moves the expected name/drink score from -5.8 to -0.7 points per guest.

1 Introduction

Speech slot filling, such as "What is your name?", is the first point of contact in most @Home tasks. Two errors cost points in opposite directions. A robot that accepts a misheard name stores wrong guest information (4×-40 in HRIC). A robot that always asks "Did you say X?" loses the non-essential-question bonus (4×15) and spends about 5 s per question. The rulebook explicitly allows asking a guest to repeat when the robot could not hear, and does not count it as a non-essential question [5]. This creates a third option that the original implementation did not use. GPSR has a different trade-off: a rephrasing request costs −30, but a misunderstood command loses both understanding (80) and solving (250).

Research question. Do calibrated STT and extraction confidence signals allow a three-way policy (accept / repeat / confirm) to reduce non-essential questions without increasing wrong memorized information, compared to the current heuristic?

Hypotheses. **H1:** the confidence features predict extraction errors with AUROC ≥ 0.80. **H2:** after calibration, ECE ≤ 0.05. **H3:** compared to the current heuristic, the calibrated policy (a) cuts confirmation questions by $\geq 50\%$, (b) with ≤ 2 percentage points more wrong memorized values, and (c) a higher expected HRIC score.

2 Related work

Modern neural networks are often over-confident, and post-hoc calibration such as temperature or Platt scaling [2, 1] or isotonic regression [6] repairs the mapping from score to probability. Whisper [3] exposes per-word probabilities that, as we show below, are not calibrated for short in-domain utterances. KnowNo [4] calibrates when an LLM planner should ask for help, using conformal prediction. We apply the same idea, "ask only when the calibrated uncertainty warrants it", to a much smaller decision with a known cost function, the competition scoresheet.

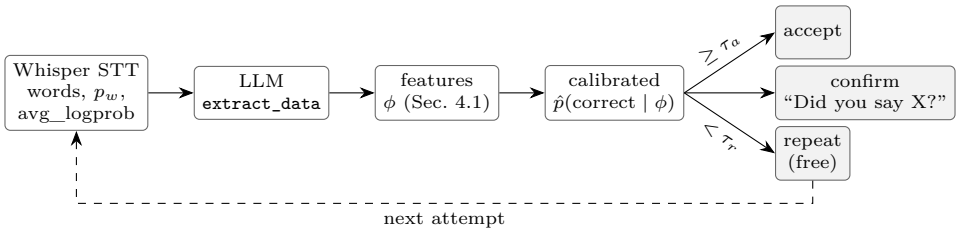


Figure 1: Decision pipeline of `HRITasks.ask_and_confirm`. Every attempt writes one JSONL record (`logs/hri_decisions/`) with the transcription, word probabilities, features, decision and the user's answer.

3 System and baseline (week 1)

Baseline. The original implementation averaged the Whisper probability of the words of the extracted value and accepted when this mean and the option similarity were both above 0.7, or when the value matched an option exactly. Otherwise it confirmed.

Normalization bug. For HRIC, name and drink are extracted by an LLM without an option list, so the similarity is always 1 and only the confidence matters. When the LLM normalizes the value ("coke" → "Coca-Cola"), none of its words appear in the transcription. The matched set is then empty, the confidence becomes 0, and the robot always confirms. In the dataset, 48 of 771 attempts (6.2%) have a normalized value, and 18 (2.3%) get a legacy confidence of exactly 0. We now fall back to the closest fuzzy span of the transcription, and to the whole transcription when no span is close enough. This changes the legacy decision in 26 attempts (3.4%), 11 of which are wrong values that the bug was "protecting". The legacy acceptance rate goes from 57.1% to 57.8%. The fix therefore removes an arbitrary failure, but it also shows that the raw Whisper score is not a safe acceptance signal by itself.

Instrumentation. `confidence_policy.py` is a ROS-free module (unit tested with mocks) that computes the features, applies the policy and writes the decision log. `CONFIDENCE_POLICY` in `hri_constants.py` selects `legacy` (default, unchanged behavior), `binary` or `three_way`. The STT microservice now also returns the segment `avg_logprob` and `no_speech_prob`.

4 Method

4.1 Features

For the extracted value v with matched word probabilities $p_1..p_k$: mean, minimum and product of p_i; mean and minimum over the whole transcription; option similarity; exact-match flag; normalized-value flag (v not a substring of the transcription); number of words of the transcription and of v; and, when available, Whisper `avg_logprob` and `no_speech_prob`. Features used: `conf_mean`, `conf_min`, `conf_prod`, `conf_whole_mean`, `conf_whole_min`, `similarity`, `exact_match`, `normalized_flag`, `n_words`, `n_target_words`, `avg_logprob`, `no_speech_prob`.

4.2 Calibration

We fit, on the calibration speakers only: isotonic regression on the mean or minimum confidence, Platt scaling on the mean, and an L_2 logistic regression on all features (with an optional isotonic layer on its out-of-fold scores). The model is selected by speaker-grouped 5-fold cross-validated log-loss on the calibration split. The held-out speakers are used only for the final numbers. We report AUROC (95% CI from a speaker-level cluster bootstrap), ECE with 10 equal-width bins, and the Brier score. The selected model is exported as JSON and evaluated on the robot by the same code (`CalibratedModel`), so no scikit-learn is needed on board.

4.3 Three-way policy

Given $\hat{p}$: accept if $\hat{p} \geq \tau_a$; ask to repeat if $\hat{p} < \tau_r$ (except on the last attempt, which confirms); otherwise confirm. The thresholds maximize the expected score per slot on the calibration split:

$$S = 15 \cdot \mathbb{1}[\text{no confirmation}] \; - 40 \cdot \mathbb{1}[\text{wrong value}] \; - 40 \cdot \mathbb{1}[\text{no value after 5 attempts}], \tag{1}$$

subject to a wrong-value rate $\leq 18.9\%$ (the current heuristic's wrong-value rate plus 2 percentage points). The cost of a missing value is our assumption: without a name or drink the later introduction items are lost. GPSR uses its own costs (-30 per rephrase, -330 per misunderstood command).

Offline replay. Data is collected with `always_confirm=True`, so every attempt has a label and the user's yes/no. A policy is replayed slot by slot. When it needs another attempt (after a "no" or a repeat), we use the next logged attempt of that slot. If there is none, we draw a random attempt of the same speaker and slot (30 replays). This counterfactual assumes that repeating gives an exchangeable new sample, which is optimistic if a speaker is consistently misheard.

5 Dataset (week 2)

Protocol: `collect_confidence_session.py` runs only the name/drink dialog with confirmation forced. For each trial it prints a target value (25% outside the official lists), which the speaker says in their own words. Speaker id and condition (accent, noise, distance) are stored in every record. Labels come from intended: 771 (a human label overrides the automatic one). Split by speaker: 29 calibration speakers (525 attempts) and 12 held-out speakers (246 attempts).

TTS proxy collection. Until the human sessions exist, `make_tts_dataset.py` follows the same protocol with synthetic speakers. Each edge-tts English voice is one speaker with a fixed condition: noise (quiet, music, crowd babble made of other voices, or fan), a base SNR of 20, 10 or 5 dB, and a distance of 0.5, 1.5 or 3 m. Distance is simulated by an exponential reverb tail (RT_{60} 0.25–0.8 s) and a lower SNR. Each attempt picks a new phrasing and speaking rate, and 60% of Coca-Cola, Sidral Mundet and orange juice targets are said as "coke", "sidral" or "OJ". The audio goes through faster-whisper with the robot settings and the hotwords and initial prompt of the HRIC dialog. The extraction LLM is replaced by rules that strip carrier phrases and normalize those aliases, exactly the case behind the normalization bug. Labels are exact because the intended value is known.

Condition	Attempts	Wrong value
native accent	483	50.9%
non-native accent	288	43.1%
0.5m	285	31.2%
1.5m	271	50.2%
SNR 5 dB	262	54.2%
SNR 20 dB	259	40.2%
SNR 10 dB	250	49.6%
crowd	237	68.4%
3.0m	215	67.4%
quiet	187	38.5%
fan	184	44.0%
music	163	33.7%

Table 1: 771 labeled attempts from 41 speakers (target ≥ 200 / ≥ 10: yes). Overall error rate 48.0% (name 47.5%, drink 48.4%). A speaker contributes to several condition rows.

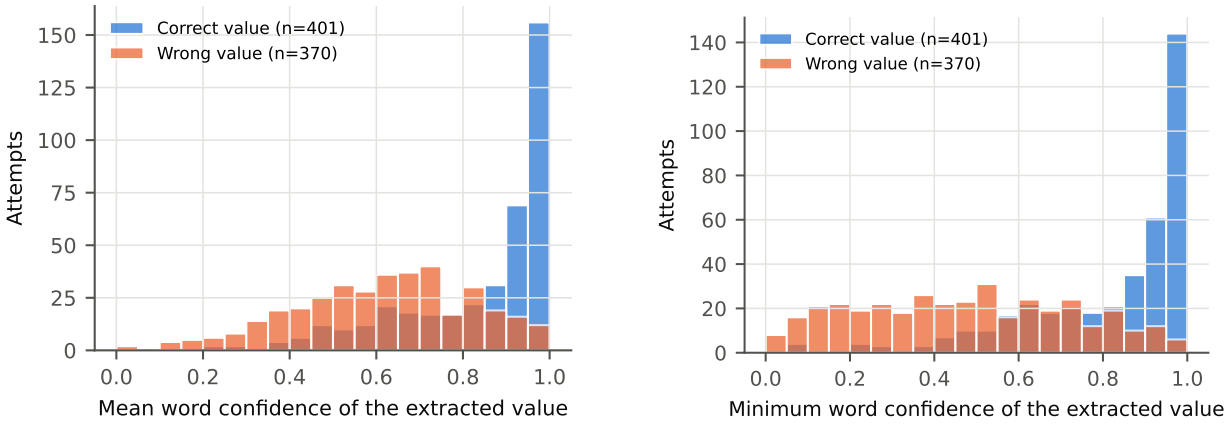


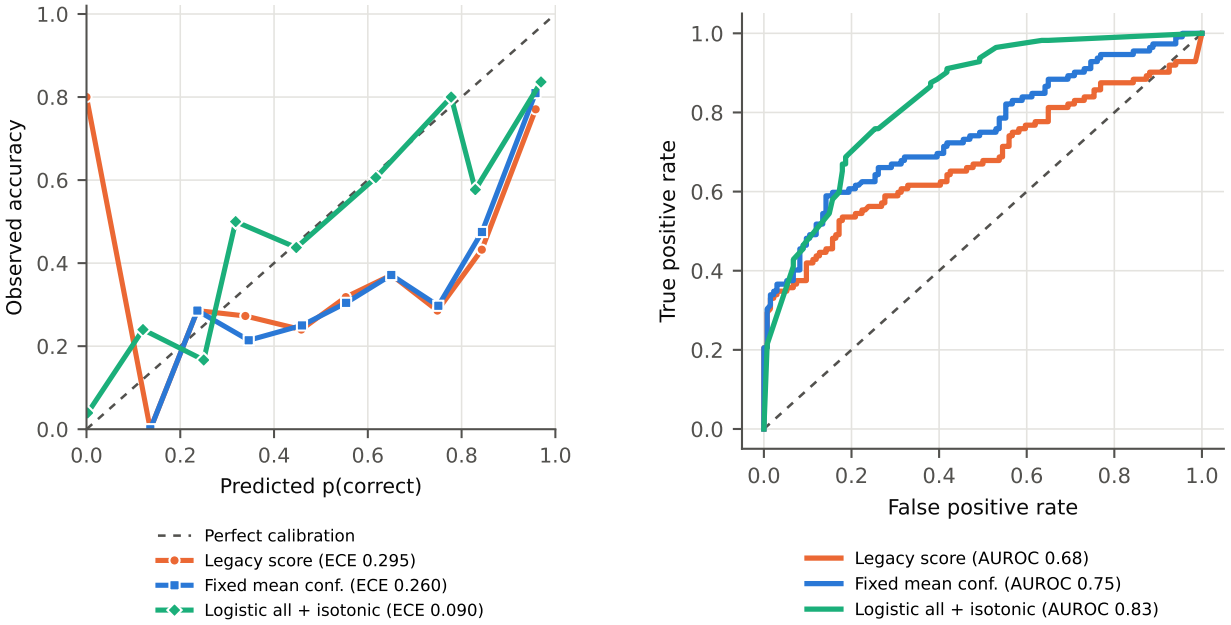
Figure 2: Whisper confidence of the extracted value for correct and wrong values. The overlap is the reason a single fixed threshold cannot separate them.

6 Results (week 3)

6.1 Calibration

Score / model (held-out speakers)	AUROC [95% CI]	ECE	Brier
Legacy score (raw)	0.68 [0.60, 0.75]	0.295	0.310
Fixed mean conf. (raw)	0.75 [0.68, 0.82]	0.260	0.275
Isotonic(mean)	0.74 [0.66, 0.81]	0.097	0.210
Isotonic(min)	0.79 [0.72, 0.85]	0.106	0.191
Platt(mean)	0.75 [0.68, 0.82]	0.120	0.217
Logistic all (C=0.1)	0.84 [0.75, 0.92]	0.093	0.171
Logistic all (C=1)	0.84 [0.75, 0.92]	0.103	0.171
Logistic all + isotonic	0.83 [0.74, 0.92]	0.090	0.176

Table 2: Error prediction on held-out speakers. “Raw” rows use the Whisper score directly as a probability. The bold model was selected on the calibration split.



(a) Reliability

(b) ROC

Figure 3: Held-out reliability diagram and ROC. The legacy score has ECE 0.295, and the fixed raw mean has ECE 0.260.

6.2 Policy comparison

Policy	Wrong	Confirm/guest	Repeat/guest	Turns/slot	s/slot	No value	Points/guest
(a) Always confirm	0.0%	4.18	0.00	4.18	23.0	11.9%	-9.5
(b) Current heuristic 0.7/0.7	28.5%	1.26	0.00	2.02	11.5	0.8%	-5.8
(b') Heuristic + matching fix	28.5%	1.20	0.00	1.99	11.4	1.3%	-4.9
(c) Calibrated binary	17.9%	1.89	0.00	2.62	14.8	8.5%	-3.2
(d) Calibrated three-way	21.8%	0.27	2.10	2.19	18.2	11.6%	-0.7

Table 3: Offline replay on held-out speakers (2 slots per guest). Thresholds: binary $\tau_a = 0.62$; three-way $\tau_a = 0.52$, $\tau_r = 0.50$. Points count only the name/drink items (+15 / −40).

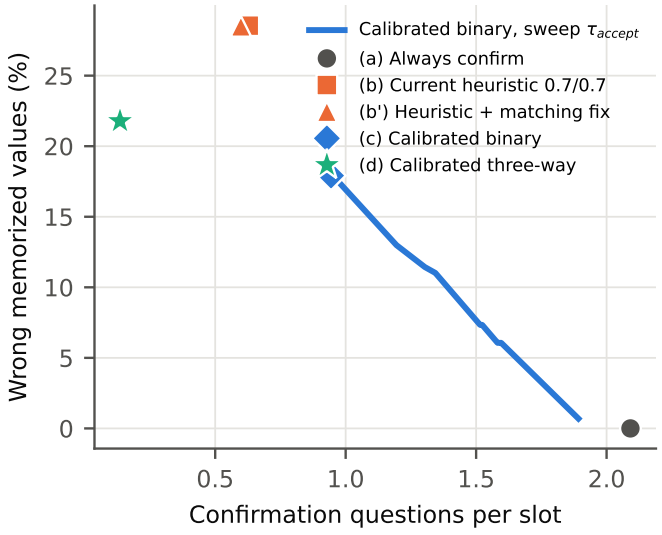


Figure 4: Trade-off between non-essential questions and wrong memorized values on held-out speakers.

7 Discussion

Hypotheses. **H1** (AUROC ≥ 0.80): supported (AUROC 0.83, CI [0.74, 0.92]). **H2** (ECE ≤ 0.05): not supported (ECE 0.090, versus 0.295 for the raw legacy score). **H3**: supported. Confirmation questions changed by 78.5% ($\geq$ 50%: yes), wrong values changed by -6.8 pp (≤ 2 pp: yes), and expected points went from -5.8 to -0.7 per guest (higher: yes). Always confirming scores -9.5.

What the scoresheet implies. With +15 for not asking and -40 for a wrong value, accepting pays off whenever $p(\text{correct}) > 40/55 \approx 0.73$, but only if p is calibrated. A raw Whisper probability of 0.7 does not mean 70% correct (Table 2). The free repeat is most useful at low $\hat{p}$, where a confirmation question would most likely be answered "no" anyway.

TTS proxy caveats. The error rate (48.0%) is far above what we expect from people at the robot: the crowd and 3 m conditions are harsh, and English TTS voices mispronounce Spanish and Yoruba targets ("sidral", "Ximena", "Oluwaseun"). Names are labeled correct when their metaphone code matches, because the robot says the name aloud (Adel / Adele, Mohammed / Muhammad); this also accepts Simone / Simon. When $\tau_a \approx \tau_r$, the three-way policy almost never confirms: a free repeat beats a -15 confirmation, at the price of more slots with no value after all attempts (Table 3). Whether people tolerate repeated "could you repeat?" prompts is a question for the live validation.

Limitations. (i) The dataset is small and synthetic in its audio. The bootstrap CIs over speakers are wide. (ii) The repeat counterfactual assumes exchangeable attempts. (iii) Whisper calibration depends on language, accent and hotwords, so the model must be refit when the STT model or prompt changes. (iv) LLM extraction errors with confident STT, such as a wrong name chosen from a sentence, are invisible to these features. This motivates the follow-up issue on LLM uncertainty (multi-sample consistency or logprobs).

8 Conclusion

On the TTS proxy data (771 attempts, 41 voices), with real Whisper probabilities, H1 is supported, H2 is not supported and H3 is supported. The calibrated three-way policy moves the expected name/drink score from -5.8 to -0.7 points per guest, with a 78.5% reduction in confirmation questions and a -6.8 pp change in wrong memorized values ($\tau_a = 0.52$, $\tau_r = 0.50$). These numbers are provisional: they show the direction of the effect with the real STT model, not its size with human speakers. The robot keeps `CONFIDENCE_POLICY = "legacy"` until the week-2 recordings confirm them.

Reproducibility. Code: `task_manager/utils/confidence_policy.py`, `research/confidence_calibration/`. `python run_all.py --data data` regenerates every number, table and figure in this document from the logs.

References

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